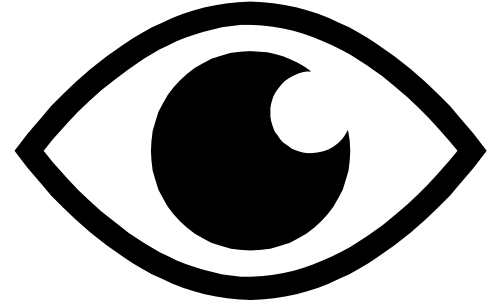


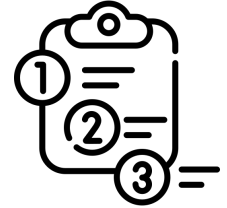


Final Presentation

D2: SightMate

by Meera Pandya, Josh Joung, Shakthi Angou

Requirements



Battery Life

minimum of **4 hours**
because a guide dog usually
takes a break every 4 hours

Accuracy (safety)

at least **70%** because it is
the minimum qualification
to become a guide dog
Aiming for **80%**

Detection Distance (safety)

minimum of **2 meters**
because a user would need
enough distance to avoid the
obstacle

Weight (health)

no more than **450 grams**,
battery pack may be
offloaded to waist if needed

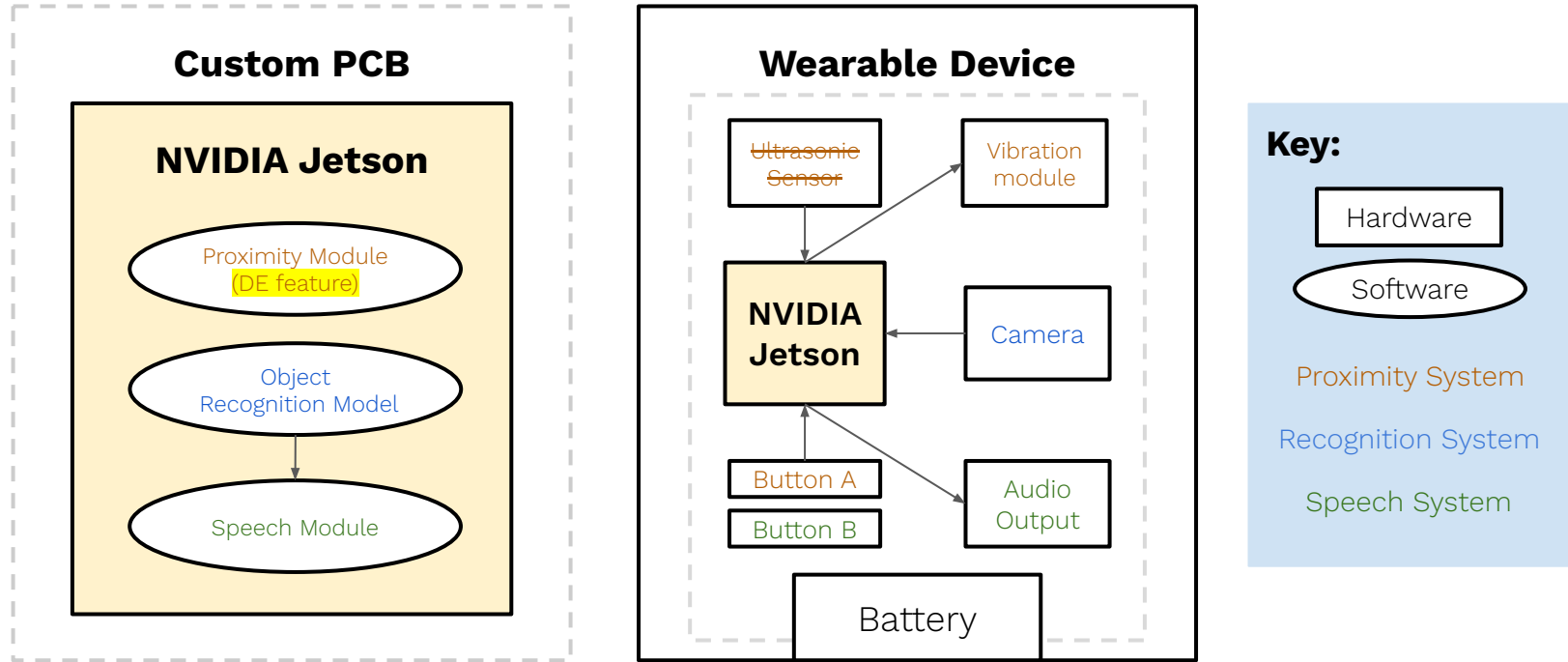
Recognition Delay (safety)

less than **2.5 seconds** to
permit 2-meter detection
(blind pedestrians walk at .8
m/s)

Noise Detection (safety)

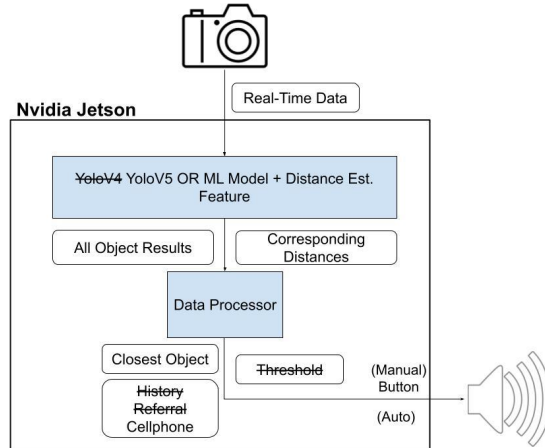
user should be able to **hear**
surrounding noises
regardless of the audio
device

Block Diagram

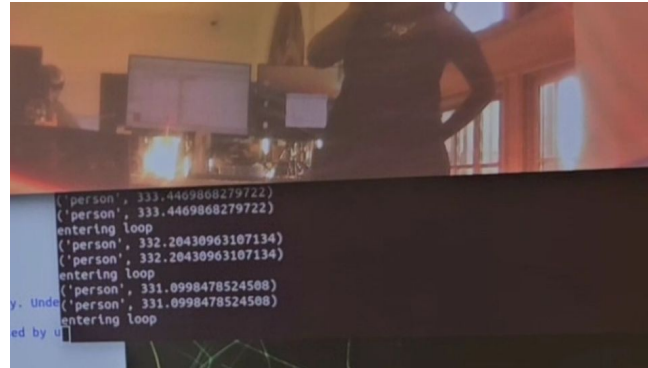


Completed: Object Recognition (OR) Module

- Use ~~YoloV4~~ **YoloV5** OR ML Model
- Created **Distance Estimation** module (using reference images)
- Use pandas to interpret reference data (.csv) for 5 objects
- Detect & est. distance every frame



```
chair_test7.jpg
[{'name': 'chair', 'boxwidth': 95.2259521484375, 'confidence': 0.86723977323468}, {'name': 'chair', 'boxwidth': 104.465576171875, 'confidence': 0.8391171097755432}, {'name': 'chair', 'boxwidth': 121.91455078125, 'confidence': 0.8363508582115173}, {'name': 'chair', 'boxwidth': 147.88623846875, 'confidence': 0.8148535490036811}, {'name': 'chair', 'boxwidth': 39.742431640625, 'confidence': 0.7098137736320496}, {'name': 'chair', 'boxwidth': 114.846923828125, 'confidence': 0.6743069291114087}, {'name': 'chair', 'boxwidth': 53.232177734375, 'confidence': 0.5866798759506775}, {'name': 'chair', 'boxwidth': 56.770751953125, 'confidence': 0.451446208537757874}, {'name': 'chair', 'boxwidth': 69.53369140625, 'confidence': 0.4455329477787018}, {'name': 'chair', 'boxwidth': 89.156982421875, 'confidence': 0.4395681917667389}, {'name': 'dining table', 'boxwidth': 291.574951171875, 'confidence': 0.27174651622772217}]
Object: chair BoxWidth: 95.2259521484375 Distance: 480.0302785187882 cm
Object: chair BoxWidth: 104.465576171875 Distance: 437.57323710945076 cm
Object: chair BoxWidth: 121.91455078125 Distance: 374.9457307524401 cm
Object: chair BoxWidth: 147.88623846875 Distance: 309.0980153266572 cm
Object: chair BoxWidth: 39.742431640625 Distance: 1150.1898209294468 cm
Object: chair BoxWidth: 114.846923828125 Distance: 398.0197188428041 cm
Object: chair BoxWidth: 53.232177734375 Distance: 858.7163305647155 cm
Object: chair BoxWidth: 56.770751953125 Distance: 805.1917362266861 cm
Object: chair BoxWidth: 69.53369140625 Distance: 657.3984410659738 cm
Object: chair BoxWidth: 89.156982421875 Distance: 512.7062299588978 cm
ref: chair_test7.jpg object: chair
```



Completed: Proximity & Speech Modules

- Device control buttons:

Control A: vibration setting (alerting user non-disruptively)

Control B: (Single Press): speech identification of immediate obstacle (MVP)

(Double Press): continuous speech identification setting

- Proximity Module:

~~Ultrasonic sensor picks up on objects within 2m of the user.~~ DE feature in the OR model is connected to the **vibration motor** placed at the back of the user's neck to alert them of obstacles approaching.

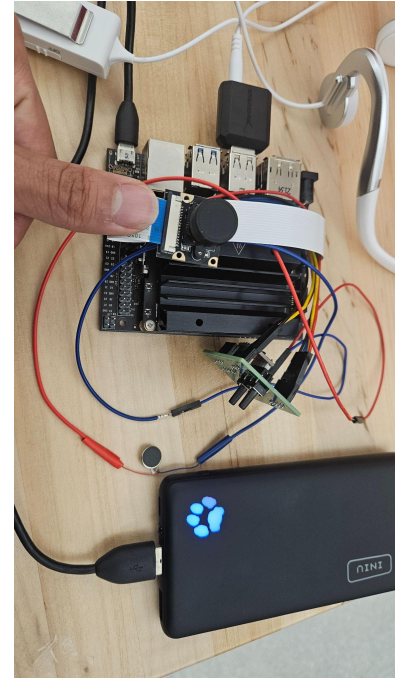
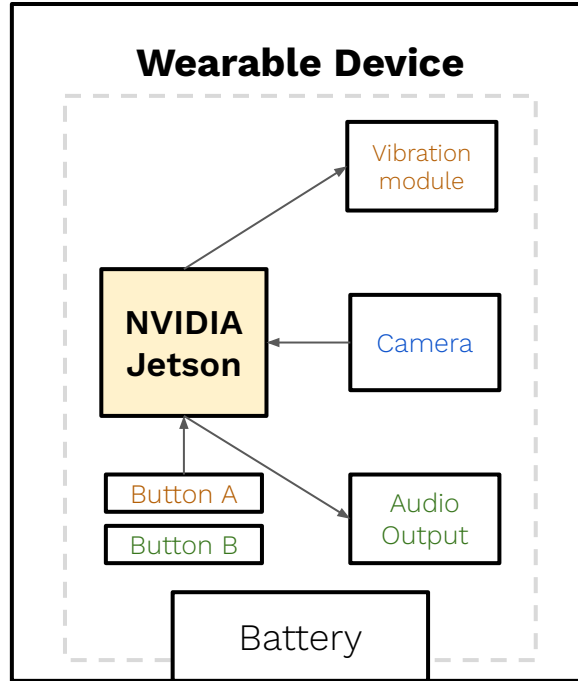
- Speech Module:

Output from the OR model is processed and converted into speech using a **TTS engine** called **espeak** that we use with the **pyttsx3** python library.

Completed: Hardware Implementation

System	Components	Integration Plan
Object Recognition System	e-CAM50_CUNX/NANO Camera	MIPI interface w/ onboard connector
Proximity Detection System	HC-SR04 Ultrasonic Sensor DE feature in OR model	Connect to Jetson GPIO pins via custom PCB for voltage conversion and/or current limiting
	Vibration Motor	
Dictation System	Control Buttons	
	CM108 Audio Converter	USB plug-in
Overall	10,000 mAh Battery	Rechargeable power bank

Complete Solution



Test, Verification and Validation

Testing	Verification	Risk Mitigation
Object recognition model	Identify the closest object using a built-in camera	Implement the model with YOLOv7 for greater accuracy
Distance estimation module	Compare the distance of the closest material measured by the model with the actual distance	
Text-to-speech module	Pass noise testing that tests whether users can hear both speech and background noise	Bone-conduction headphones
Vibration module	Vibrate if there is an object within 2m in front of the user	Recalibrate the distance estimation feature
Device controls (buttons)	Turns on and off (vibration module, auto/manual settings) when the user presses the button	Unit-testing for the different modes
Module integration	-Compare the time for the product to provide the result to the minimal recognition delay. -The overall product weighs less than 450g -Battery lasts more than 4 hours	Find bottlenecks in our system to improve latency
Functionality	The device detects and alerts the closest object within 2m of range from the user	All of the above

Test Success

Testing	Metrics	Result
Object Recognition Model	> 70% on identifying an object	95% (38/40 images, 5 objects)
Distance Estimation Feature	$\pm 30\text{cm}$ of actual object distance	Tests done on 4 different distances. Average of 21.5% uncertainty within $\pm 30\text{cm}$
Text-to-speech Module	user-testing for surrounding sounds 20 trials each object	100% (20/20 person, 20/20 couch, 20/20 chair, 20/20 cat, 20/20 cellphone)
Vibration Module	> 95% accuracy on vibration	100% (20/20 on person, 20/20 on nothing)
Device Controls (buttons)	100% accuracy on controls	100% (20/20 on button A, 20/20 on button B)
Module Integration (weight)	< 450g on the overall product weight	192g (device) + 209g (battery) = 401g < 450g

Test Failure & Improvement Plan

Testing	Metrics	Failure	Plan
Recognition Delay	< 2.5s to recognize an object	~8 seconds delay for 20 seconds testing. Frame delay due to the latency of the OR model	Reduce the bottleneck of the model -restrict the number of classes to identify) -skip a detection every 3 frames
Battery Life	> 4 hours of running	Headless device needs to be deployed	Deploy the device headless and run it on the battery
Functionality	> 70% on the accuracy	Headless device needs to be deployed	Measure accuracy with the headless device with LAMP volunteers

Trade-offs

- Chose **pre-trained model** instead of trained model with indoor object dataset

Model	Real Objects	Detected	Falsely detected	Percentage (%)
Pre-trained	58	49	5	84.4
Trained	58	21	4	36.2

- Chose **Distance Estimation feature** in the OR model instead of ultrasonic sensor
 - Ultrasonic sensor does not work well with Jetson Nano
 - DE feature rarely goes over $\pm 30\text{cm}$, although **some calibration is necessary**

Actual (m)	Detected (m)	Off (m)
1.80	1.82	+ 0.02
1.20	0.89	- 0.31
0.20	0.38	+ 0.18
2.2	1.94	- 0.26

Timeline

Workload Split:

Hardware by **Meera**

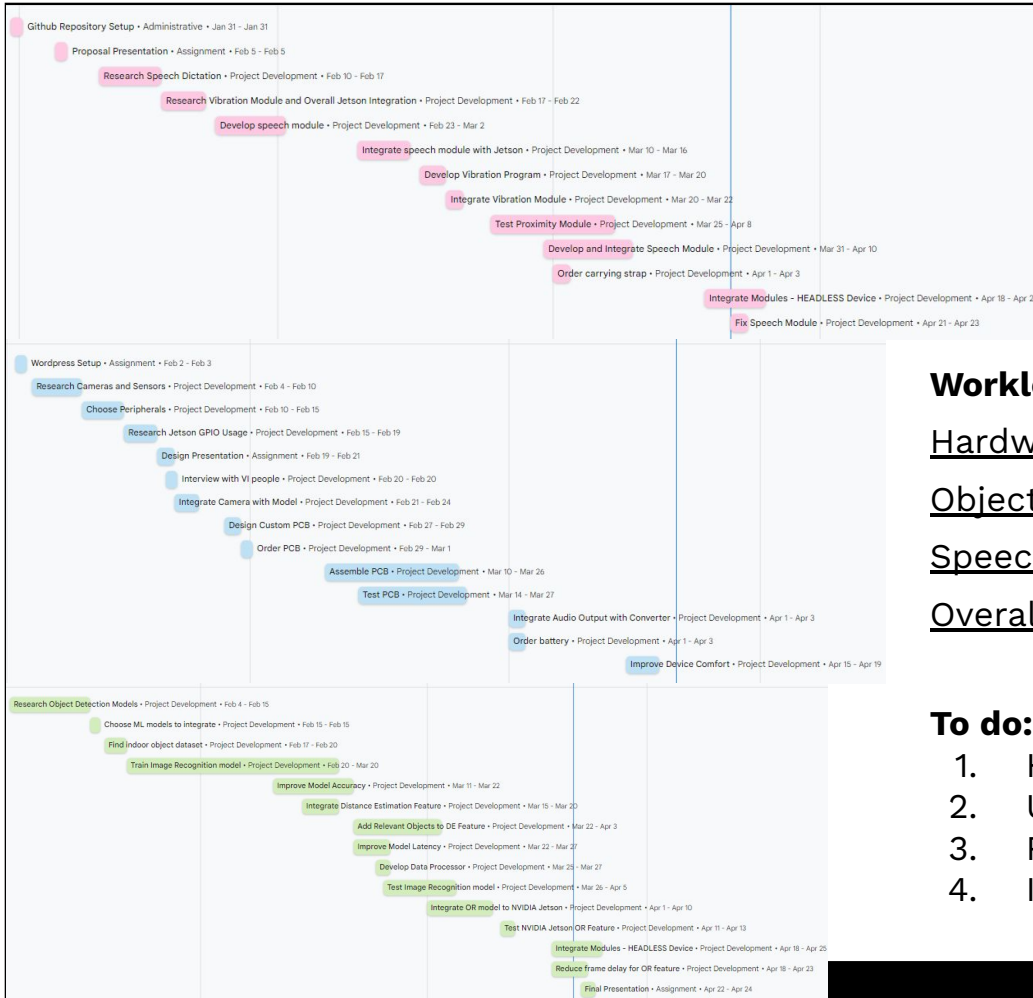
Object Recognition Model by **Josh**

Speech and Vibration Modules by **Shakthi**

Overall Integration and Device Design by **All**

To do:

1. Headless Device
2. User Testing
3. Reduce frame delay
4. Improve product appearance



Project Takeaways

Technical knowledge/technology necessary for the project

1. Machine Learning techniques, frameworks (Josh)
2. Circuit & PCB design, GPIO configuration (Meera)
3. Linux Sound Architecture, Sensor & GPIO Configuration (Shakthi)
4. NVIDIA Jetson Nano configuration/programming (All)

Importance of slack time

1. Integration of multiple modules took longer than expected
2. Progress not met as expected (conflict with other courses or events)

Reading documentation / Research

1. Github README & Issues
2. NVIDIA tutorials
3. Stack Overflow
4. YouTube videos